Quiz 2 Solutions, MA2031, Oct 9, 2023

1. Let $A = \begin{bmatrix} -3 & -2 & 0 & -1 \\ 8 & 5 & 1 & 2 \\ 7 & 4 & 2 & 1 \\ -2 & -1 & -1 & 0 \end{bmatrix}$. Construct bases for

   (a) $R(A)$ and (b) $N(A)$. [5]

   Solution: $RREF(A) = \begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. [2]

   Basis for $R(A)$ is $\left\{ \begin{bmatrix} -3 \\ 8 \\ 7 \\ -2 \end{bmatrix}, \begin{bmatrix} -2 \\ 5 \\ 4 \\ -1 \end{bmatrix} \right\}$. [1]

   Basis for $N(A)$ is $\left\{ \begin{bmatrix} 2 \\ -3 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 0 \\ -1 \end{bmatrix} \right\}$. [2]

2. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 1 & 2 \end{bmatrix}$. Consider the ordered bases

   $C = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$ and $E = \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$ for $\mathbb{R}^{3\times 1}$.

   Compute the coordinate matrix $[A]_{E,C}$. [5]

   (Note: $[A]_{E,C}$ is the matrix that satisfies $[Ax]_E = [A]_{E,C}[x]_C$ for all $x \in \mathbb{R}^{3\times 1}$.)

   Solution: We compute the RREF of the augmented matrix with its columns as $w_1, w_2, w_3, Av_1, Av_2, Av_3$, where v_1, v_2, v_3 are vectors in C and w_1, w_2, w_3 are vectors in E. Now,

   $Av_1 = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}$, $Av_2 = \begin{bmatrix} 3 \\ 6 \\ 6 \end{bmatrix}$, $Av_3 = \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix}$. [1]

   $$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 2 & 3 & 2 \\ -1 & 0 & 1 & 4 & 6 & 4 \\ 0 & 1 & -1 & 5 & 6 & 4 \end{array}\right] \xrightarrow{RREF} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 7 & 9 & 6 \\ 0 & 1 & 0 & 16 & 21 & 14 \\ 0 & 0 & 1 & 11 & 15 & 10 \end{array}\right].$$ [3]

   So, $[A]_{E,C} = \begin{bmatrix} 7 & 9 & 6 \\ 16 & 21 & 14 \\ 11 & 15 & 10 \end{bmatrix}$. [1]

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3. Compute the QR-factorization of the matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}. \qquad [6]$$

Solution: Gram-Schmidt orthonormalization of the columns of A yield the vectors

$$\begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 0 \\ 1/\sqrt{3} \end{bmatrix}, \quad \begin{bmatrix} -1/\sqrt{15} \\ 2/\sqrt{15} \\ 3/\sqrt{15} \\ -1/\sqrt{15} \end{bmatrix}, \quad \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \\ 0 \end{bmatrix}. \qquad [3]$$

Then, $Q = \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{15} & 1/\sqrt{3} \\ 1/\sqrt{3} & 2/\sqrt{15} & -1/\sqrt{3} \\ 0 & 3/\sqrt{15} & 1/\sqrt{3} \\ 1/\sqrt{3} & -1/\sqrt{15} & 0 \end{bmatrix}$. [1]

Then, $R = Q^*A = \dfrac{1}{\sqrt{3}} \begin{bmatrix} 3 & 1 & 2 \\ 0 & \sqrt{5} & -\sqrt{5} \\ 0 & 0 & 3 \end{bmatrix}$. [2]

4. Determine all least squares solutions of the system of linear equations:

$x + z = 6, \quad x + y - z = 0, \quad x + 2y - 3z = 0.$ [4]

Solution: The system is $AX = b$, where $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & 2 & -3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$.

We then have

$$A^*A = \begin{bmatrix} 3 & 3 & -3 \\ 3 & 5 & -7 \\ -3 & -7 & 11 \end{bmatrix}, \quad A^*b = \begin{bmatrix} 6 \\ 0 \\ 6 \end{bmatrix}. \qquad [1]$$

$$[A^*A|A^*b] \xrightarrow{RREF} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 5 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right]. \qquad [2]$$

Thus, the least squares solutions are given by

$$\begin{bmatrix} 5 \\ -3 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix} \quad \text{for } \alpha \in \mathbb{R}. \qquad [1]$$